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IBNet: Interactive Branch Network for salient object detection

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ABSTRACT

At present, saliency detection methods have achieved gratifying progress benefiting from the development of deep learning. However, the existing methods always fail to make full use of the label information. To address this problem, we focus on the complementarity of salient body information and salient detail information within the labels and propose the Interactive Branch Network (IBNet) in this paper. Generally, IBNet contains three components of Label Redefinition Module (LRM), Information Exchange Module (IEM) and Connected Flow Loss (CFL). These three components all play an enormously important role in the complementary performance of detection. In LRM, enough useful and meaningful heuristic knowledge from the given labels is expanded for dynamic and collaborative supervised learning. In IEM, different derived branches are assigned to collect different types of features for interactive fusion. In CFL, the losses from all connected branches are merged to calculate the total loss. Extensive experiments on benchmark datasets exhibit the effectiveness and efficiency of the proposed method against the state-of-the-art approaches. The source code is publicly available at https://github.com/xianfangfx/IBNet.

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1. Introduction

Salient object detection is an active research topic, which is attracting increasing attention. It aims to capture the most visually distinctive regions of the given image or video. This technology has a widespread application prospect in image retrieval [1], video compression [2], semantic segmentation [3], visual tracking [4], robot navigation [5] and other computer vision fields. More introductions can be found in recent survey papers [6,7].

During the past decade, many traditional hand-crafted saliency detection methods [8–10] have been proposed, which are heavily dependent on limited prior knowledge such as color, texture and appearance. Up to now, due to the development of deep learning such as Convolutional Neural Networks (CNNs) [11] and Fully Convolutional Networks (FCNs) [12], a large number of modern saliency detection methods [13–25] have emerged, which could be roughly divided into two categories, namely, aggregation-based methods and edge-based methods. Typically, the former obtain the saliency map by first extracting different levels of features with the encoder and then integrating these features with the decoder, while the latter take saliency mask as supervision with the help of edge label prior to obtain the saliency map. Once the good saliency map is obtained, the desired detection effect can be achieved.

Although remarkable progresses have been made, there still remain some challenges for algorithms in promoting detection performance. For example, it has long been regarded as a primary bottleneck to deal with the detection of complex scenes with blurred, occluded, illuminated, rotated, and various scale objects. Apart from that, pixels close to the edge are more difficult to be detected than those near the center to a certain extent, and it is hence also a major bottleneck to handle the imbalance reasonably. Motivated by LDF [26], we observe the fact that the aforementioned problems can be well alleviated by making full use of label information.

On the basis of this observation, we are committed to producing labels with rich information from known saliency mask. For several examples with different scale objects, the illustrate of our defined labels are shown in Fig. 1. In fact, these labels belong to the extension of either body label or detail label, in which the body label reduces the dispersion of pixels from the edge to the center and the detail label is beneficial to reserving the edge characteristics. Each label processed from the mask is potentially associated with others, since they share the similarity with the original label. The dissimilarity between them lies in that the body label concentrates on the feedback of pixel information in the center, while the detail

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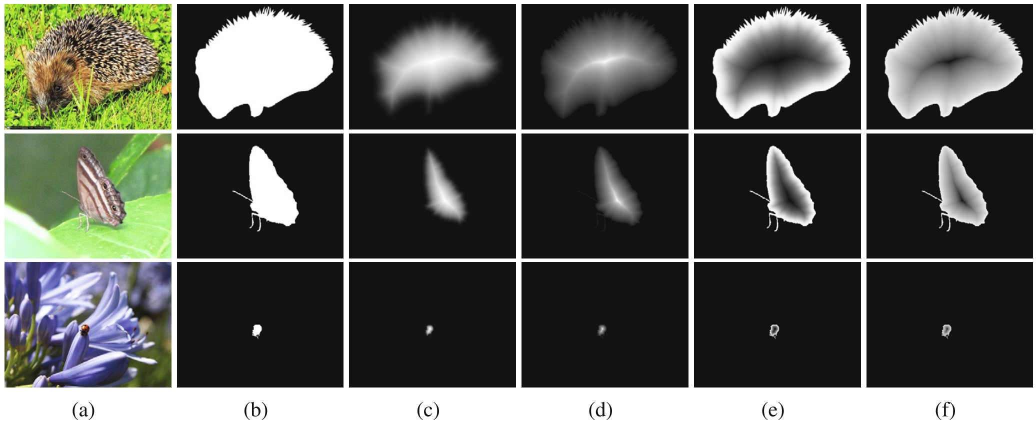

Fig. 1. Illustrate of our defined labels on several examples with different scale objects. (a) Image; (b) Ground truth; (c) Strengthened body label; (d) Weakened body label; (e) Strengthened detail label; (f) Weakened detail label.

label concentrates on the feedback of pixel information in the edge. After receiving the pure body label and detail label, the specific information will be immediately strengthened or weakened to tackle the intractable situation in an adaptive way, thereby generating the strengthened body label, weakened body label, strengthened detail label and weakened detail label. In consequence, these practical labels are adequately differentiated, and the diversity of information can be guaranteed.

Different from the previous methods, we attempt to focus on the complementarity of features from the generated advanced labels. For this purpose, a novel Interactive Branch Network (IBNet) is proposed in this paper. IBNet contains three components of Label Redefinition Module (LRM), Information Exchange Module (IEM) and Connected Flow Loss (CFL). These three components all play an enormously important role in the complementary performance of detection. Compared to LDF, IBNet has many unique advantages. For example, LDF only generates two types of labels and lacks sufficient label information, while IBNet more concretely explores the pixels worthy of special treatment, thereby generating four different types of labels with more abundant label information. Also, in LDF, the body label and detail label generated by the mask of each image are constant and the distinguishability of these labels is very finite, while in IBNet, it can adaptively coordinate the body and detail parts, so as to get the body labels and detail labels which are gradually changeable strengthened and weakened and these labels are very distinguishable. To intuitively reveal the advantages, some heat maps of the labels produced by LDF and IBNet are shown in Fig. 2. The labels generated by our method have more discriminative centers and edges than those generated by the comparison method. Beside, the mask is equivalent to adding the body label to the detail label, and the rule is also universal to the strengthened and weakened body labels as well as detail labels.

Specifically, we first peel the saliency mask into two parts of the pseudo body label and pseudo detail label according to the position factors between all foreground pixels and the nearest background pixel, and then design two smooth distribution functions to flexibly act on the pseudo body and pseudo detail labels, so as to generate the strengthened body, weakened body, strengthened detail and weakened detail labels. Furthermore, we establish structural branches corresponding to four types of labels and learn lots of heuristic knowledge of salient body information and salient detail information dynamically and cooperatively through the information interaction of complementary features in two phases. After that, we calculate the sum of the losses of all iterations from the connected branches and obtain the final saliency map.

Our contributions are summarized as follows:

- We discuss the implicit correlation between the body label and detail label, and produce the appropriate labels with more information than existing labels.
- We propose the IBNet for salient object detection by explicitly exploring and refining multiple cues of sufficient salient body information and salient detail information.

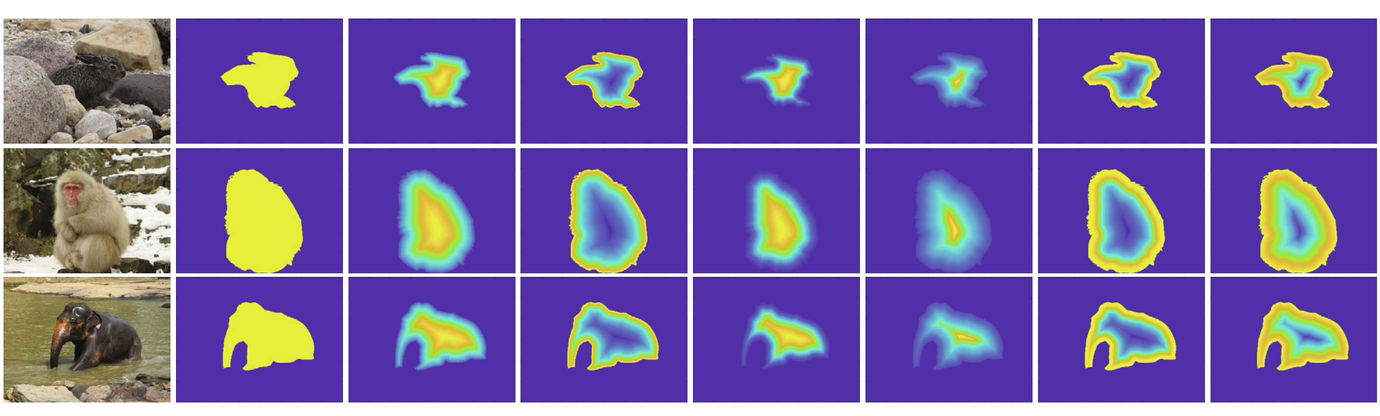

Fig. 2. Some heat maps of the labels produced by LDF and IBNet. (a) Image; (b) Ground truth; (c-d) Labels of body and detail obtained from LDF, respectively; (e-h) Labels of strengthened body, weakened body, strengthened detail and weakened detail obtained from IBNet, respectively.

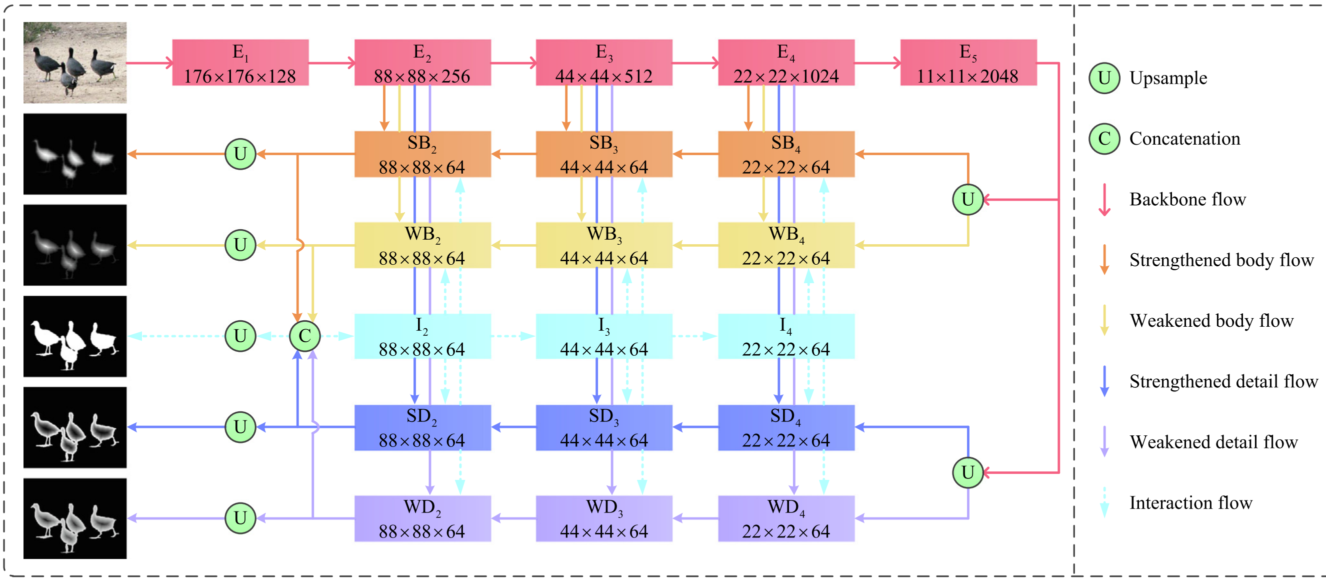

Fig. 3. The overall framework of the proposed model. Our model is based on ResNet-50 [37]. $\{E_i\}_{i=1}^5$ and $\{I_i\}_{i=2}^4$ are two encoders, while $\{SB_i\}_{i=2}^4$, $\{WB_i\}_{i=2}^4$, $\{SD_i\}_{i=2}^4$ and $\{WD_i\}_{i=2}^4$ are four decoders.

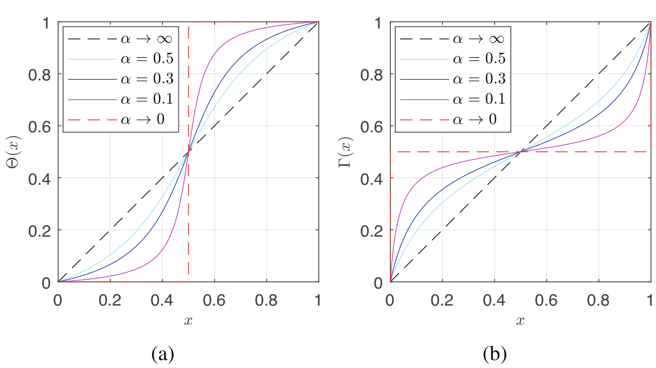

Fig. 4. Two smooth distribution functions. (a) The curved concave-convex function; (b) The curved convex-concave function.

- We compare the proposed method with 17 state-of-the-arts approaches on five datasets in terms of five evaluation metrics to demonstrate its effectiveness and efficiency.

2. Related work

We briefly review the work related to salient object detection in two aspects, which include accuracy saliency detection and real-time saliency detection.

2.1. Accuracy saliency detection

The detection accuracy of saliency object is constantly pursued by researchers. In the feature extraction network, convolution layers of different levels often have different degrees of feature representation ability. On the one hand, multi-level feature fusion is able to greatly improve detection accuracy. On the other hand, incorporating attention mechanisms into networks is also a popular way to improve detection accuracy.

For instance, Huo et al. [27] proposes an object-level saliency detection model by leveraging bottom-up attention and objectness cues. Li et al. [28] proposes a deep contrast network, which combines a pixel-level fully convolutional stream and a segment-wise spatial pooling stream. Liu et al. [29] proposes a two-stage network, which first automatically learns cues of global contrast, objectiveness, compactness and their optimal combination, and then progressively corrects the saliency map by integrating local context information. Xiang et al. [30] proposes a saliency bias and diffusion method to mine the spatial support of salient objects, which is dedicated to the learning of saliency confidences. Zhang et al. [31] proposes a bi-directional message passing model to integrate multi-level features. Ji et al. [32] proposes a graph based bottom-up salient object detection model, which fuses saliency maps using low-level features and objectness features under a manifold ranking model. However, these methods can not fully excavate label information and feed these underlying information to feature network to obtain powerful features.

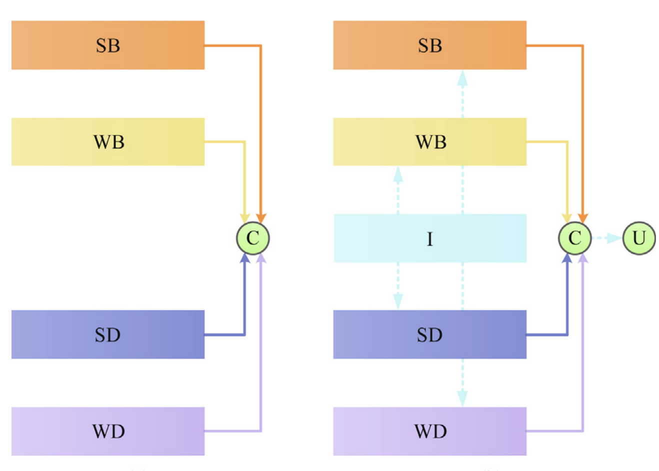

Fig. 5. Two phase procedures. (a) The first phase procedure; (b) The second phase procedure.

2.2. Real-time saliency detection

In addition to the accuracy, the real-time detection of salient object is also the interest of researchers. The features at higher layers usually have context-aware information and are suitable for locating the center regions. The features at lower layers usually have spatial information and are suitable for locating the edge regions. To accelerate the detection, it is valuable to abandon the low-level features, since its high-resolution processing seriously restricts the detection speed.

For instance, Lee et al. [33] proposes a unified deep learning model, which encodes high features at low resolution and can be reused in each query area. Xi et al. [34] proposes a simplified end-to-end neural network, which can quickly predict a dense full-resolution saliency map for the image with compact pipelines. Li et al. [35] proposes a dual branch network, in which the perceptual branch and objectness branch can obtain features simultaneously. Wang et al. [36] proposes a network with boundary information guidance, in which two separate decoding sub-networks have a saliency sub-network and a boundary sub-network to learn saliency and boundary information. However, these methods do not catch into the importance of label information and use them in low-level features or high-level features.

3. Our method

We propose the Interactive Branch Network (IBNet) to detect salient objects, which contains three main components of Label Redefinition Module (LRM), Information Exchange Module (IEM) and Connected Flow Loss (CFL). The overall network framework is shown in Fig. 3.

3.1. Label Redefinition Module

In order to expand enough useful and meaningful heuristic knowledge from the given labels, Label Redefinition Module (LRM) is proposed.

It is assumed that $f \in \mathscr{F}$ and $b \in \mathscr{B}$ denote the foreground pixel and background pixel, respectively. In a binary image, for each pixel p that has its original pixel value l, we get the distance value as the modified pixel value d from the nearest background pixel by the distance transformation, which can be written as:

$$d = \begin{cases} \min\limits_{b \in \mathscr{B}} dist(p, b), & p \in \mathscr{F} \\ 0, & p \notin \mathscr{F} \end{cases}, \quad (1)$$

where $dist(\cdot)$ is the Euclidean distance between paired of pixels. The modified pixel d is mapped further to get the normalized pixel value l' using the min–max scaling, which can be written as:

$$l' = \frac{d' - \min(d')}{\max(d') - \min(d')}, \quad (2)$$

where $d' = \sqrt{d}$ and the value range is constrained in the interval $[0, 1]$. As a result, the values of the pixels near the center of the object are relatively large, while those close to the edge of the object are relatively small. We directly let l' be the pseudo body label and indirectly subtract it from l to get the pseudo detail labels, which can be written as:

$$\begin{cases} l_{\mathrm{PB}} = l' \\ l_{\mathrm{PD}} = l - l' \end{cases}, \quad (3)$$

where l_{PB} and l_{PD} denote the pseudo body label and pseudo detail label, respectively. Additionally, for each new pixel value l', we design two smooth distribution functions to map it again to redefine diverse labels in view of the distribution of all pixels. The functions $\Theta(x)$ and $\Gamma(x)$ are defined as:

$$\begin{cases} \Theta(x) = \frac{2}{\pi\beta} \arctan\left(\frac{2}{\alpha}\left(x - \frac{1}{2}\right)\right) + \frac{1}{2} \\ \Gamma(x) = \frac{\alpha}{2} \tan\left(\frac{\pi\beta}{2}\left(x - \frac{1}{2}\right)\right) + \frac{1}{2} \end{cases}, \quad (4)$$

where $\beta = (4/\pi)\arctan(1/\alpha)$ and $\alpha \in (0, \infty)$ is a tuning parameter. The designed smooth distribution functions are shown in Fig. 4. When α infinitely tends to 0, $\Theta(x)$ and $\Gamma(x)$ with respect to x are close to horizontal line and vertical line, respectively. When α tends to ∞, both $\Theta(x)$ and $\Gamma(x)$ with respect to x are close to raw line. Moreover, the two functions also hold the intrinsic properties as follows:

- They are symmetric about the line $y = x$ with the same parameter value.
- Both of them are centrosymmetric about the coordinate point (0.5, 0.5).
- Both of them are evenly scattered in the coordinate intervals [0, 0.5) and (0.5, 1].

Without loss of generality, α is set to 0.3. Besides, we multiply these resulting values by the normalized pixel value and original pixel value to eliminate the interference of background and diffuse the influence of the foreground, forming the strengthened body, weakened body, strengthened detail and weakened detail labels, which can be written as:

$$\begin{cases} l_{\mathrm{SB}} = l \times l' \times \Theta(l') \\ l_{\mathrm{WB}} = l \times l' \times \Gamma(l') \\ l_{\mathrm{SD}} = l - l \times l' \times \Theta(l') \\ l_{\mathrm{WD}} = l - l \times l' \times \Gamma(l') \end{cases}, \quad (5)$$

where $l_{\mathrm{SB}}, l_{\mathrm{SD}}, l_{\mathrm{WB}}$ and l_{WD} denote the strengthened body label, weakened body label, strengthened detail label and weakened detail label, respectively. After redefining the label, the information of abundant labels will be obtained for supervised learning.

3.2. Information exchange module

Feature fusion from different pyramid levels is the key to boost network capability. Similarly, the fusion of different derived branch features is also crucial in enhancing network capability. To this end, Information Exchange Module (IEM) is proposed.

Suppose there is an input image with size $H \times W$ in a framework that essentially conforms to the encoder-decoder mode. As for the encoder features $\widetilde{\mathrm{E}} = \left\{\widetilde{\mathrm{E}}_1, \widetilde{\mathrm{E}}_2, \widetilde{\mathrm{E}}_3, \widetilde{\mathrm{E}}_4, \widetilde{\mathrm{E}}_5\right\}$, they can be easily extracted from the given backbone network through the encoder $\{\mathrm{E}_i\}_{i=1}^5$, corresponding to the number of channels of $\{128, 256, 512, 1024, 2048\}$. Here, we just take features $\left\{\widetilde{\mathrm{E}}_2, \widetilde{\mathrm{E}}_3, \widetilde{\mathrm{E}}_4, \widetilde{\mathrm{E}}_5\right\}$, since feature $\widetilde{\mathrm{E}}_1$ brings a considerable time overhead with a negligible performance improvement. As for the four types of decoder features $\widetilde{\mathrm{SB}} = \left\{\widetilde{\mathrm{SB}}_2, \widetilde{\mathrm{SB}}_3, \widetilde{\mathrm{SB}}_4\right\}, \widetilde{\mathrm{WB}} = \left\{\widetilde{\mathrm{WB}}_2, \widetilde{\mathrm{WB}}_3, \widetilde{\mathrm{WB}}_4\right\}, \widetilde{\mathrm{SD}} = \left\{\widetilde{\mathrm{SD}}_2, \widetilde{\mathrm{SD}}_3, \widetilde{\mathrm{SD}}_4\right\}$ and $\widetilde{\mathrm{WD}} = \left\{\widetilde{\mathrm{WD}}_2, \widetilde{\mathrm{WD}}_3, \widetilde{\mathrm{WD}}_4\right\}$, they are distinguished by the branches of four flows through the encoders $\{\mathrm{SB}_i\}_{i=2}^4, \{\mathrm{WB}_i\}_{i=2}^4, \{\mathrm{SD}_i\}_{i=2}^4$ and $\{\mathrm{WD}_i\}_{i=2}^4$, and uniformly squeezed to 64 channel numbers. Among them, the two branches of strengthened body flow and weakened body flow are responsible for the collection of body features, while the other two branches of strengthened detail flow and weakened detail flow are responsible for the collection of detail features. As for the encoder features $\widetilde{\mathrm{I}} = \left\{\widetilde{\mathrm{I}}_2, \widetilde{\mathrm{I}}_3, \widetilde{\mathrm{I}}_4\right\}$, they are aggregated by four

Table 1
Quantitative comparison of different methods on five datasets in terms of five evaluation metrics of F_{max}, F_{avg}, E_m, S_m and MAE. The larger the value of F_{max}, F_{avg}, E_m and S_m, the better the performance. The smaller the value of MAE, the better the performance. The first best, the second best and the third best results are highlighted in red, green and blue, respectively. '–' means the authors have not provided corresponding saliency maps. '†' means using pre-processing or post-processing. '-R', '-R34', '-R101' and '-X101' mean using ResNet-50, ResNet-34, ResNet-101 and ResNeXt-101 [38] as the backbone network, respectively. If not specified, VGG-16 [39] is used as the backbone network.

Method	ECSSD					DUT-OMRON					PASCAL-S					HKU-IS					DUTS-TE				
	F_{max}	F_{avg}	E_m	S_m	MAE	F_{max}	F_{avg}	E_m	S_m	MAE	F_{max}	F_{avg}	E_m	S_m	MAE	F_{max}	F_{avg}	E_m	S_m	MAE	F_{max}	F_{avg}	E_m	S_m	MAE
Amulet	0.915	0.869	0.912	0.893	0.059	0.743	0.647	0.784	0.780	0.098	0.837	0.768	0.829	0.820	0.098	0.897	0.842	0.914	0.884	0.051	0.778	0.671	0.798	0.803	0.085
DSS†	0.916	0.900	0.924	0.882	0.053	0.772	0.729	0.846	0.788	0.066	0.836	0.804	0.851	0.797	0.096	0.900	0.856	0.927	0.880	0.050	0.825	0.789	0.885	0.824	0.056
R^3Net†-X101	0.934	0.914	0.940	0.910	0.040	0.795	0.748	0.859	0.817	0.062	0.843	0.803	0.845	0.805	0.094	0.915	0.894	0.945	0.895	0.035	–	–	–	–	–
PAGR	0.927	0.894	0.917	0.889	0.061	0.771	0.711	0.843	0.775	0.071	0.856	0.807	0.854	0.818	0.093	0.919	0.887	0.941	0.889	0.047	0.854	0.784	0.883	0.838	0.056
PiCANet-R	0.935	0.886	0.927	0.917	0.046	0.803	0.717	0.848	0.832	0.065	0.870	0.804	0.862	0.854	0.076	0.919	0.870	0.941	0.905	0.044	0.860	0.759	0.873	0.869	0.051
BANet-R	0.945	0.923	0.953	0.924	0.035	0.803	0.746	0.865	0.832	0.059	0.875	0.835	0.887	0.852	0.070	0.930	0.899	0.955	0.913	0.032	0.872	0.815	0.907	0.879	0.040
SCRN-R	0.950	0.918	0.942	0.927	0.037	0.811	0.746	0.869	0.837	0.056	0.887	0.837	0.887	0.868	0.064	0.935	0.897	0.954	0.917	0.033	0.887	0.808	0.901	0.885	0.040
EGNet-R	0.947	0.920	0.947	0.925	0.037	0.815	0.756	0.874	0.841	0.053	0.875	0.828	0.877	0.852	0.075	0.935	0.901	0.956	0.918	0.031	0.889	0.815	0.907	0.887	0.039
PAGE†	0.931	0.906	0.943	0.912	0.042	0.791	0.736	0.860	0.824	0.062	0.857	0.814	0.878	0.839	0.078	0.920	0.884	0.948	0.904	0.036	0.838	0.777	0.886	0.854	0.052
AFNet	0.935	0.908	0.941	0.913	0.042	0.797	0.739	0.859	0.826	0.057	0.868	0.826	0.886	0.849	0.071	0.925	0.889	0.949	0.906	0.036	0.863	0.792	0.895	0.867	0.046
CPD-R	0.939	0.917	0.949	0.918	0.037	0.797	0.747	0.873	0.825	0.056	0.869	0.829	0.887	0.847	0.072	0.925	0.891	0.952	0.906	0.034	0.865	0.805	0.904	0.869	0.043
PoolNet-R	0.949	0.919	0.948	0.926	0.035	0.805	0.752	0.874	0.831	0.054	0.890	0.837	0.888	0.866	0.065	0.936	0.903	0.959	0.919	0.030	0.889	0.819	0.912	0.886	0.037
CapSal-R101	0.862	0.825	0.866	0.826	0.074	0.639	0.564	0.703	0.674	0.096	0.866	0.825	0.878	0.838	0.073	0.884	0.843	0.907	0.850	0.058	0.823	0.755	0.866	0.815	0.062
BASNet-R34	0.942	0.880	0.921	0.916	0.037	0.805	0.756	0.869	0.836	0.056	0.862	0.779	0.852	0.837	0.077	0.930	0.898	0.947	0.908	0.033	0.859	0.791	0.884	0.866	0.048
F^3Net-R	0.945	0.925	0.946	0.924	0.033	0.813	0.766	0.876	0.838	0.053	0.883	0.845	0.893	0.860	0.063	0.937	0.909	0.959	0.917	0.028	0.891	0.840	0.918	0.888	0.035
MINet-R	0.947	0.924	0.953	0.925	0.033	0.810	0.755	0.873	0.833	0.055	0.879	0.840	0.898	0.856	0.064	0.935	0.908	0.961	0.920	0.028	0.884	0.828	0.917	0.884	0.037
LDF-R	0.949	0.927	0.951	0.923	0.035	0.819	0.768	0.879	0.838	0.054	0.884	0.847	0.899	0.860	0.062	0.941	0.914	0.961	0.919	0.028	0.896	0.842	0.923	0.890	0.035
IBNet-R (ours)	0.951	0.929	0.952	0.927	0.032	0.817	0.776	0.880	0.836	0.052	0.886	0.850	0.901	0.862	0.060	0.941	0.918	0.961	0.922	0.027	0.898	0.855	0.930	0.892	0.034

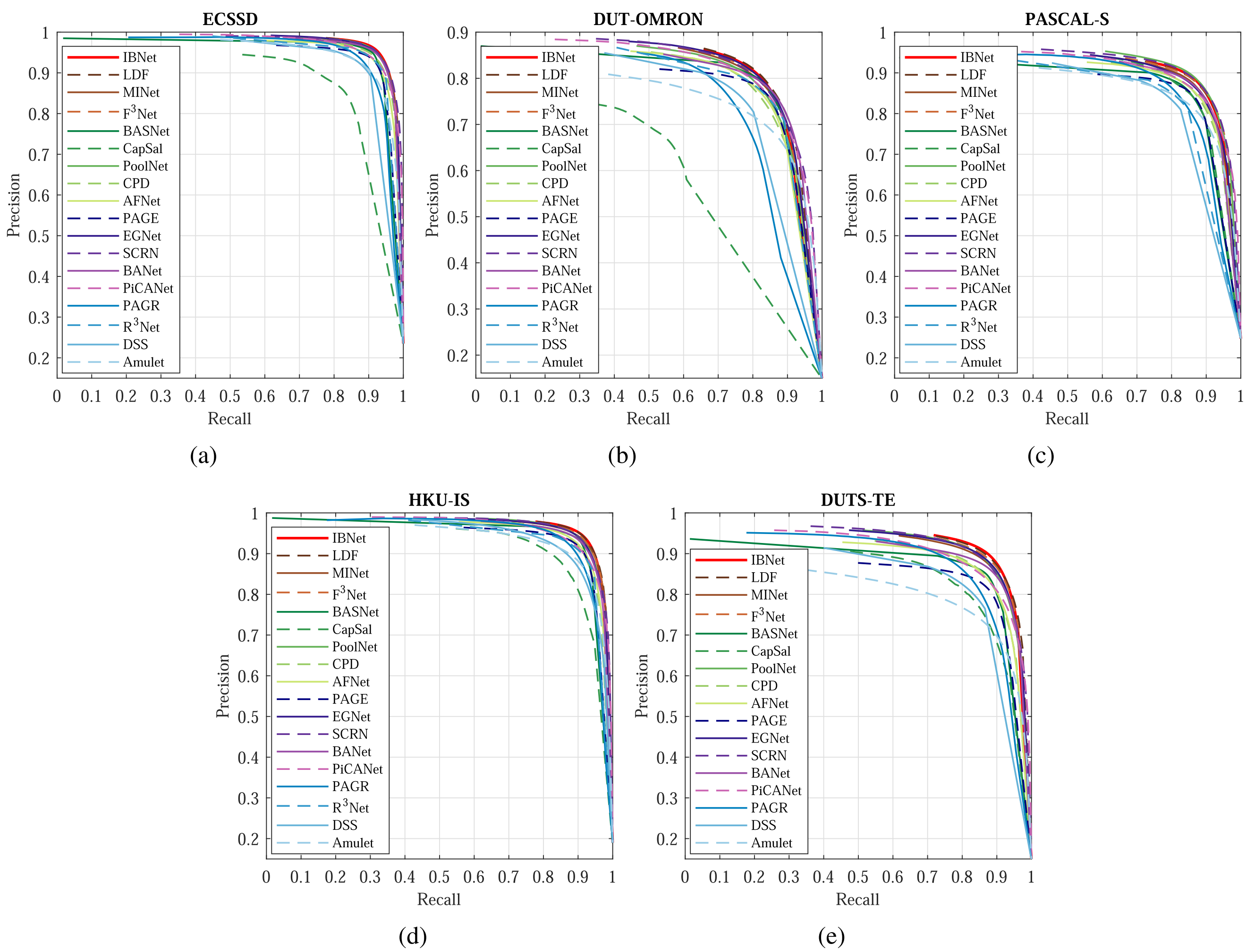

Fig. 6. The PR curves of different methods on five datasets. The larger the area under the PR curves, the better the performance. (a) ECSSD; (b) DUT-OMRON; (c) PASCAL-S; (d) HKU-IS; (e) DUTS-TE.

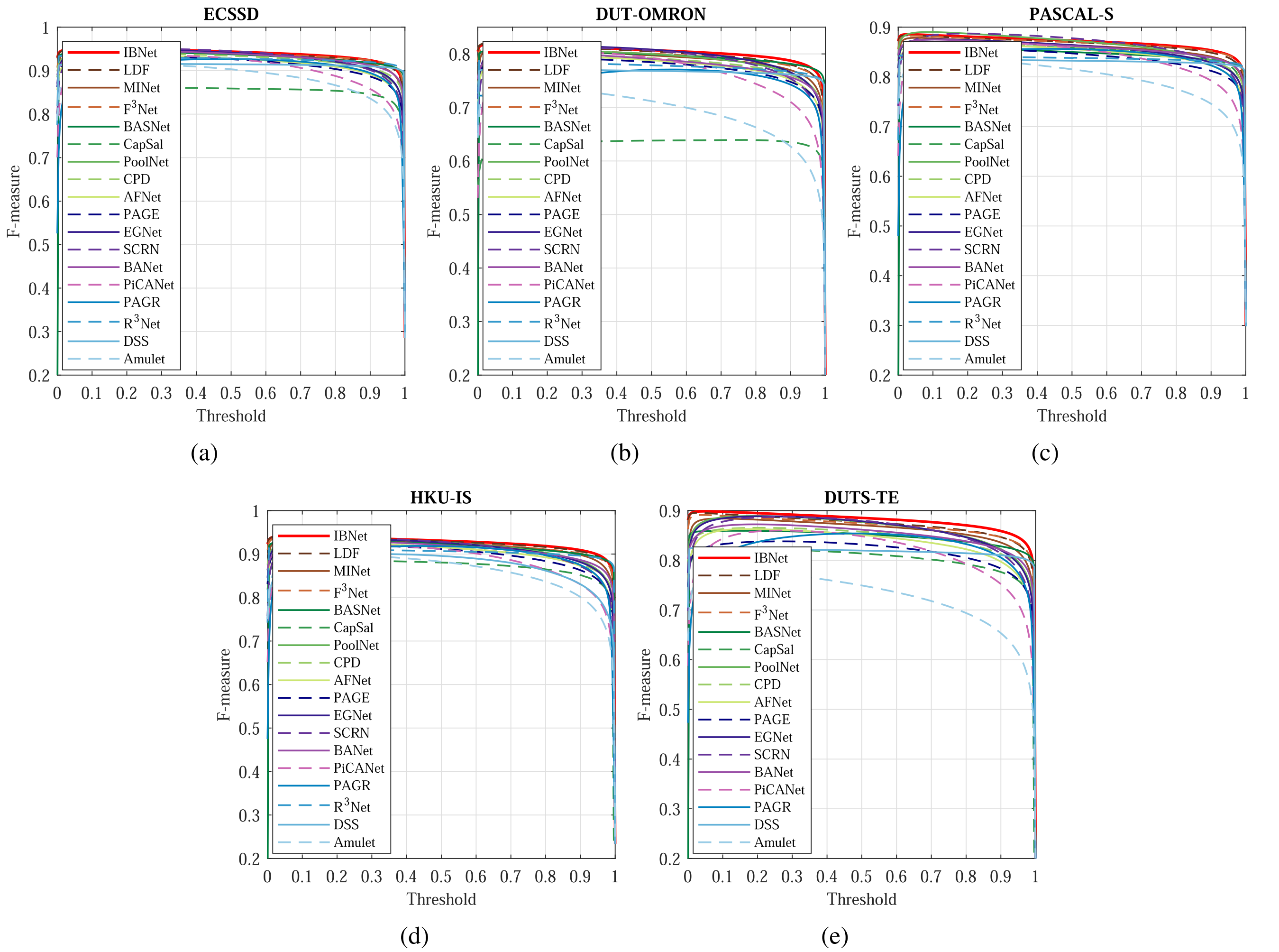

Fig. 7. The F_β curves of different methods on five datasets. The larger the area under the F_β curves, the better the performance. (a) ECSSD; (b) DUT-OMRON; (c) PASCAL-S; (d) HKU-IS; (e) DUTS-TE.

connected branches through the encoder $\{\mathrm{I}_i\}_{i=2}^{4}$ for exchanging information. These four branches happen in parallel not sequentially, the complementary of that makes the feature interaction particularly intimate, thus is capable of obtaining the excellent saliency map. To enable a favorable flow of the features across the framework, we make intermediate connections, which can be classified into two phases. Actually, the first phase corresponds to the first iteration, while the second phase corresponds to the second iteration or more iterations. Two phase procedures are shown in Fig. 5. These two procedures are completely independent of each other. To avoid being too complicated, the number of iterations in the second phase is set to 1 just as in the first phase. For input feature $\widetilde{f}^{(0)}$, the two phase procedures P_1 and P_2 can be expressed mathematically as:

$$P_1 \Rightarrow \begin{cases} \widetilde{\mathrm{SB}}^{(1)} = \mathrm{SB}\left(\widetilde{f}^{(0)}\right) \\ \widetilde{\mathrm{WB}}^{(1)} = \mathrm{WB}\left(\widetilde{f}^{(0)}\right) \\ \widetilde{\mathrm{SD}}^{(1)} = \mathrm{SD}\left(\widetilde{f}^{(0)}\right) \\ \widetilde{\mathrm{WD}}^{(1)} = \mathrm{WD}\left(\widetilde{f}^{(0)}\right) \\ \widetilde{f}^{(1)} = Concat\left(\widetilde{\mathrm{SB}}^{(1)}, \widetilde{\mathrm{WB}}^{(1)}, \widetilde{\mathrm{SD}}^{(1)}, \widetilde{\mathrm{WD}}^{(1)}\right) \end{cases}, \qquad (6)$$

and

$$P_2 \Rightarrow \begin{cases} \widetilde{\mathrm{SB}}^{(2)} = \mathrm{SB}\left(\widetilde{f}^{(0)}, \mathrm{I}\left(\widetilde{f}^{(1)}\right)\right) \\ \widetilde{\mathrm{WB}}^{(2)} = \mathrm{WB}\left(\widetilde{f}^{(0)}, \mathrm{I}\left(\widetilde{f}^{(1)}\right)\right) \\ \widetilde{\mathrm{SD}}^{(2)} = \mathrm{SD}\left(\widetilde{f}^{(0)}, \mathrm{I}\left(\widetilde{f}^{(1)}\right)\right) \\ \widetilde{\mathrm{WD}}^{(2)} = \mathrm{WD}\left(\widetilde{f}^{(0)}, \mathrm{I}\left(\widetilde{f}^{(1)}\right)\right) \\ \widetilde{f}^{(2)} = UP\left(Concat\left(\widetilde{\mathrm{SB}}^{(2)}, \widetilde{\mathrm{WB}}^{(2)}, \widetilde{\mathrm{SD}}^{(2)}, \widetilde{\mathrm{WD}}^{(2)}\right)\right) \end{cases}, \qquad (7)$$

where the superscript numbers stands for the number of iterations, $\widetilde{f}^{(1)}$ and $\widetilde{f}^{(2)}$ denote output features in the first and second phases, respectively, $Concat(\cdot)$ and $Up(\cdot)$ are the concatenation operation and the upsample operation, respectively. It is worth pointing out that the framework does not involve the exchange of information in the first iteration until the second iteration.

3.3. Connected flow loss

All connected branches should contribute a certain degree of loss to the whole subject such that they can complement each other in turn. Relying on this idea, Connected Flow Loss (CFL) denoted as $\mathscr{L}_{CF}$ is proposed.

For the sake of simplicity, we formalize the losses caused by the task of the strengthened body, weakened body, strengthened

Fig. 8. Qualitative comparison of different methods on several representative examples. (a) Image; (b) Ground truth; (c) IBNet; (d) LDF; (e) PoolNet; (f) CPD; (g) AFNet; (h) EGNet.

detail, weakened detail flows along with the segmentation task in the kth iteration as $\mathscr{L}_{\mathrm{SB}}^{(k)}, \mathscr{L}_{\mathrm{WB}}^{(k)}, \mathscr{L}_{\mathrm{SD}}^{(k)}, \mathscr{L}_{\mathrm{WD}}^{(k)}$ and $\mathscr{L}_{\mathrm{Seg}}^{(k)}$, respectively. Therefore, $\mathscr{L}_{\mathrm{CF}}$ is defined as:

$$\mathscr{L}_{\mathrm{CF}} = \frac{1}{2}\sum_{k=1}^{2}\sum_{\eta\in\{\mathrm{SB,WB,SD,WD}\}}\left(\mathscr{L}_{\eta}^{(k)} + \lambda\mathscr{L}_{\mathrm{Seg}}^{(k)}\right), \tag{8}$$

where λ is a hyperparameter that balances the relationship of the two kinds of loss items. By default, it is set to 1 to treat both tasks equally. Specifically, basic Binary Cross Entropy Loss (BCEL) denoted as $\mathscr{L}_{\mathrm{BCE}}$ is utilized for calculating $\mathscr{L}_{\mathrm{SB}}, \mathscr{L}_{\mathrm{WB}}, \mathscr{L}_{\mathrm{SD}}$ and $\mathscr{L}_{\mathrm{WD}}$, which can be written as:

$$\mathscr{L}_{\mathrm{BCE}} = -\sum_{p\in\mathscr{P},g\in\mathscr{G}}(g\log p + (1-g)\log(1-p)), \tag{9}$$

where $\mathscr{P}$ and $\mathscr{G}$ denote the saliency map and ground truth, respectively. In addition, standard Intersection over Union (IoU) loss denoted as $\mathscr{L}_{\mathrm{IoU}}$ together with BCEL are utilized for calculating $\mathscr{L}_{\mathrm{Seg}}$, which can be written as:

$$\mathscr{L}_{\mathrm{IoU}} = 1 - \frac{\sum_{p\in\mathscr{P},g\in\mathscr{G}}(p\times g)}{\sum_{p\in\mathscr{P},g\in\mathscr{G}}(p+g-p\times g)}. \tag{10}$$

Note that BCEL considers the local structure and ignores the global structure, while IoU loss takes them into account at the same time. Nevertheless, we utilize BCEL rather than IoU loss to cope with the four branch flows, because their labels are not binary and only meets the requirements of BCEL but not IoU loss.

4. Experiments

First of all, we describe the experimental setup including the datasets, evaluation metrics and implementation details. Next, we report the performance comparison between our method and

Table 2
Ablation study on the DUTS-TE dataset. The best results are highlighted in red.

Baseline	LRM	IEM	CFL	F_{max}	F_{avg}	E_m	S_m	MAE
✓				0.829	0.738	0.859	0.842	0.057
✓	✓			0.852	0.813	0.878	0.862	0.044
✓	✓	✓		0.886	0.836	0.901	0.884	0.038
✓	✓	✓	✓	0.898	0.855	0.930	0.892	0.034

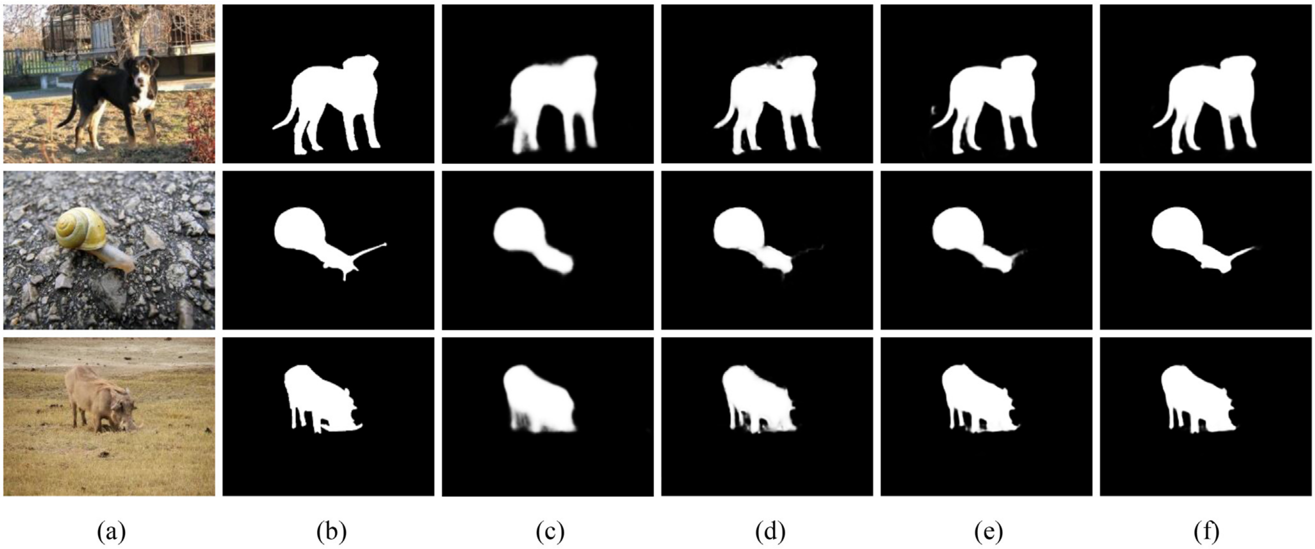

Fig. 9. Visual comparison of the impacts of individual component along with their combinations of our method on several examples. (a) Image; (b) Ground truth; (c) Baseline; (d) Baseline+LRM; (e) Baseline+LRM+IEM; (f) Baseline+LRM+IEM+CFL.

the state-of-the-art approaches on quantitative and qualitative experiments. Last but not least, we give the ablation study of the proposed individual component along with their combinations.

4.1. Experimental setup

Datasets. Five commonly used benchmark datasets including ECSSD [40], DUT-OMRON [41], PASCAL-S [42], HKU-IS [43] and DUTS [44] are selected. ECSSD, DUT-OMRON, PASCAL-S and HKU-IS consist of 1000, 5168, 850 and 4447 images, respectively. DUTS is composed of the training set with 10553 images (i.e., DUTS-TR) and the testing set with 5019 images (i.e., DUTS-TE). Various models are trained on the training set from the last one dataset and tested on its test set along with the first four datasets. It should be pointed out that only 1447 images are used in HKU-IS by us following [13,43,45].

Evaluation Metrics. Five widely used evaluation metrics including maximum F-measure (F_{max}) [46], average F-measure (F_{avg}) [46], E-measure (E_m) [47], S-measure (S_m) [48] and Mean Absolute Error (MAE) [49] are chosen. Note that F_β is defined as the weighted harmonic mean of precision and recall. F_{max} and F_{avg} are maximum and average value calculated from F_β, respectively. E_m estimates similarity by combining the local pixel values with the image-level mean value. S_m measures region-aware and object-aware structural similarities between the predicted saliency map and the ground truth saliency map. MAE reflects the pixel-wise absolute difference within the predicted and ground truth saliency maps. In addition, we also plot the Precision-Recall (PR) curves and F-measure (F_β) curves.

Implementation Details. All experiments are implemented using a GeForce RTX 2080 Ti GPU based on PyTorch repository. As for the data augmentation strategy, there are horizontal flipping, random cropping and multi-scale conversion. ResNet-50 [37] is employed as the backbone network and Stochastic Gradient Descent (SGD) is adopted for optimizing. Our model is coarsely trained and finely trained for 48 epochs with a batch size of 32, a weight decay of 5e-4, an initial learning rate of 0.05, a momentum of 0.9. The input images are resized to 352$\times$352.

4.2. Comparison with state-of-the-arts

We compare our IBNet with other 17 state-of-the-arts, which are Amulet [50], DSS [45], R^3Net [51], PAGR [52], PiCANet [53], BANet [54], SCRN [55], EGNet [56], PAGE [57], AFNet [58], CPD [59], PoolNet [60], CapSal [61], BASNet [62], F^3Net [63], MINet [64] and LDF [26]. The saliency maps of these methods are provided by the authors or computed by their released codes with default settings for fair comparisons.

Quantitative comparison. The comparison results of different methods on five datasets in terms of five evaluation metrics of F_{max}, F_{avg}, E_m, S_m and MAE are listed in Table 1. As can be seen that our IBNet is highly competitive compared to the others on all datasets in terms of these evaluation metrics. In particular, our method can obtain the first best result in most cases, which is averagely improved by 0.7% over the second best method F^3Net, and 0.9% over the third method LDF on all evaluation metrics with respect to the MAE. Besides, IBNet can achieve the top three results in almost all cases. The PR curves and F_β curves of different methods on five datasets are shown in Fig. 6 and Fig. 7, respectively. It can be seen that IBNet outperforms others at most thresholds whether with respect to the PR curves or F_β curves. The curves by our method are especially outstanding, which is attributed to the gradual refinement of information obtained through complementary branches.

Qualitative comparison. The comparison results of different methods on several representative examples are shown in Fig. 8. These examples comprehensively cover a vast range of challenging scenarios, like small scale, middle scale and large scale objects, or one, two and even more objects, or shaded, cluttered background and low contrast foreground objects. Obviously, compared with other competitors, IBNet consistently separates out more exact and satisfactory saliency maps from the original images. The saliency maps of IBNet extraordinarily approximates to the ground truth, which can not only grab the coherent boundaries, but also suppress the redundant noises. Most importantly, our method achieves these results without any pre-processing and post-processing.

4.3. Ablation study

We conduct a series of ablation studies to verify the impact of each component of our method.

The results of ablation study on the DUTS-TE dataset are listed in Table 2. The baseline refers to Feature Pyramid Networks (FPNs) [65]. Different schemes are prepared including baseline, baseline +LRM, baseline+LRM+IEM and baseline+LRM+IEM+CFL. We separately install the components on the baseline. When we use the scheme of baseline+LRM, we only generate auxiliary labels without extra fusing and the total branch is still two branches instead of four branches. When we use the scheme of baseline+LRM+IEM, we get four branches, but only the losses of two main branches are calculated. When we use the scheme of baseline+LRM+IEM +CFL, we calculate the losses of four complete branches. These results suggest that each scheme achieves significant performance improvement over baseline. Meanwhile, the results of the combinations of all components are far better than others. The comparison results of the impacts of individual component along with their combinations of our method on several examples are shown in Fig. 9. It is clear that the scheme of baseline+LRM+IEM+CFL performs the best in all the schemes.

5. Conclusion

In this paper, we investigate the complementarity issue of salient body information and salient detail information within the labels to propose the Interactive Branch Network (IBNet) for salient object detection. Three core components of Label Redefinition Module (LRM), Information Exchange Module (IEM) and Connected Flow Loss (CFL) are embedded in IBNet. Accordingly, LRM extends enough useful and meaningful heuristic knowledge from the given labels for dynamic and collaborative supervised learning, IEM assigns different derived branches to collect different types of features for interactive fusion, and CFL merges the losses from all connected branches to calculate the total loss. All the three components permeate, interfere and guide each other, which constitute the entirety of IBNet. Extensive experiments on benchmark datasets exhibit that the proposed method is superior to the state-of-the-art approaches.

CRediT authorship contribution statement

Xian Fang: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization. **Jinchao Zhu:** Data curation, Writing - review & editing, Visualization. **Ruixun Zhang:** Writing - review & editing. **Xiuli Shao:** Writing - review & editing. **Hongpeng Wang:** Writing - review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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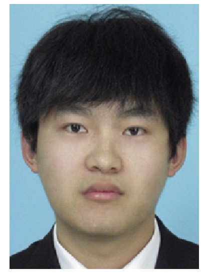

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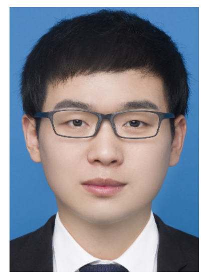

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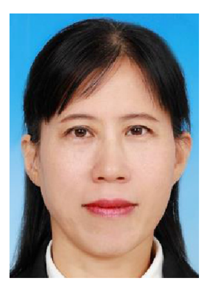

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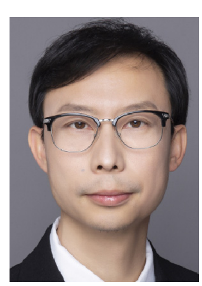

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